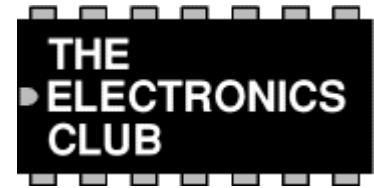


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Integrated Circuits (Chips)



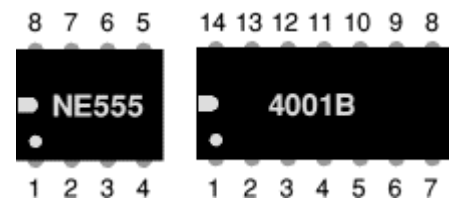
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Also see: [4000 Series ICs](#) | [74 Series ICs](#) | [555 and 556 Timer Circuits](#)

Integrated Circuits are usually called ICs or chips. They are complex circuits which have been etched onto tiny chips of semiconductor (silicon). The chip is packaged in a plastic holder with pins spaced on a 0.1" (2.54mm) grid which will fit the holes on stripboard and breadboards. Very fine wires inside the package link the chip to the pins.

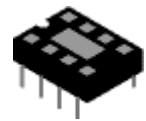
Pin numbers

The pins are numbered anti-clockwise around the IC (chip) starting near the notch or dot. The diagram shows the numbering for 8-pin and 14-pin ICs, but the principle is the same for all sizes.



Chip holders (DIL sockets)

ICs (chips) are easily damaged by heat when soldering and their short pins cannot be protected with a heat sink. Instead we use a chip holder, strictly called a DIL socket (DIL = Dual In-Line), which can be safely soldered onto the circuit board. The chip is pushed into the holder when all soldering is complete.



Chip holders are only needed when soldering so they are not used on breadboards.

Commercially produced circuit boards often have chips soldered directly to the board without a chip holder, usually this is done by a machine which is able to work very quickly. Please don't attempt to do this yourself because you are likely to destroy the chip and it will be difficult to remove without damage by de-soldering.

Removing a chip from its holder

If you need to remove a chip it can be gently prised out of the holder with a small flat-blade screwdriver. Carefully lever up each end by inserting the screwdriver blade between the chip and its holder and gently twisting the screwdriver. Take care to start lifting at both ends before you attempt to remove the chip, otherwise you will bend and possibly break the pins.

Static precautions

Many ICs are **static sensitive** and can be damaged when you touch them because your body may have become charged with static electricity, from your clothes for example. Static sensitive ICs will be supplied in antistatic packaging with a warning label and they should be left in this packaging until you are ready to

use them.

It is usually adequate to earth your hands by touching a metal water pipe or window frame before handling the IC but for the more sensitive (and expensive!) ICs special equipment is available, including earthed wrist straps and earthed work surfaces. You can make an earthed work surface with a sheet of aluminium kitchen foil and using a crocodile clip to connect the foil to a metal water pipe or window frame with a $10k\Omega$ resistor in series.



Antistatic bags for ICs
Photograph © [Rapid Electronics](#)

Datasheets

Datasheets are available for most ICs giving detailed information about their ratings and functions. In some cases example circuits are shown. The large amount of information with symbols and abbreviations can make datasheets seem overwhelming to a beginner, but they are worth reading as you become more confident because they contain a great deal of useful information for more experienced users designing and testing circuits.

PDF files

To view and print PDF files you need an Acrobat Reader which may be downloaded free for [Windows](#), [Mac](#), [RISC OS](#), or [UNIX/Linux](#) computers. If you are not sure which type of computer you have it is probably Windows.

Datasheets are available as PDF files from:

- [DatasheetArchive.com](#)
- [Datasheets.org.uk](#)
- [DatasheetCatalog.com](#)

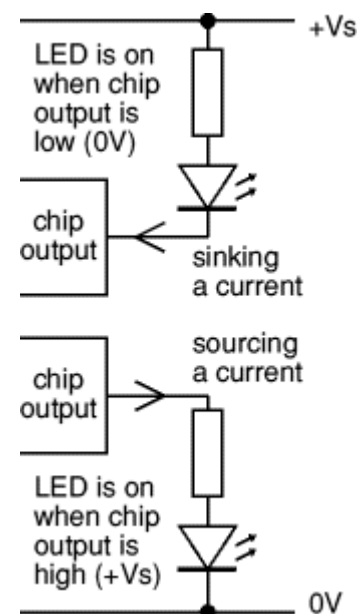
Sinking and sourcing current

Chip outputs are often said to 'sink' or 'source' current. The terms refer to the direction of the current at the chip's output.

If the chip is **sinking current** it is flowing **into the output**. This means that a device connected between the positive supply ($+V_s$) and the chip output will be switched **on when the output is low (0V)**.

If the chip is **sourcing current** it is flowing **out of the output**. This means that a device connected between the chip output and the negative supply (0V) will be switched **on when the output is high ($+V_s$)**.

It is possible to connect two devices to a chip output so that one is on when the output is low and the other is on when the output is high. This arrangement is used in the [Level Crossing](#) project to make the red LEDs flash alternately.



The maximum sinking and sourcing currents for a chip output are usually the same but there are some exceptions, for example [74LS TTL](#) logic chips can sink up to 16mA but only source 2mA.

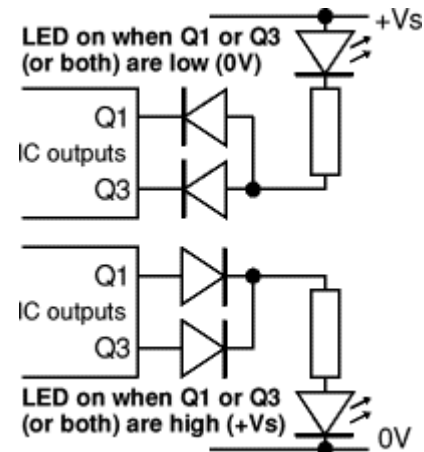
Using diodes to combine outputs

The outputs of chips (ICs) must never be directly connected together. However, diodes can be used to combine two or more digital (high/low) outputs from a chip such as a counter. This can be a useful way of producing simple logic functions without using logic gates!

The diagram shows two ways of combining outputs using diodes. The diodes must be capable of passing the output current. 1N4148 signal diodes are suitable for low current devices such as LEDs.

For example the outputs Q0 - Q9 of a [4017](#) 1-of-10 counter go high in turn. Using diodes to combine the 2nd (Q1) and 4th (Q3) outputs as shown in the bottom diagram will make the LED flash twice followed by a longer gap. The diodes are performing the function of an [OR gate](#).

Example projects: [Traffic Light](#) | [Dice](#) | [Model Lighthouse](#)

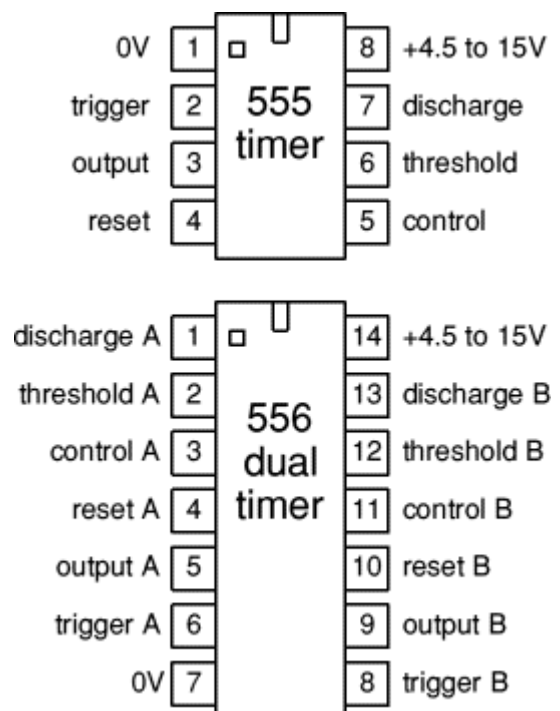


The 555 and 556 Timers

The 8-pin 555 timer chip is used in many projects, a popular version is the NE555. Most circuits will just specify '555 timer IC' and the NE555 is suitable for these. The 555 output (pin 3) can [sink and source](#) up to 200mA. This is more than most chips and it is sufficient to supply LEDs, relay coils and low current lamps. To switch larger currents you can [connect a transistor](#).

The 556 is a dual version of the 555 housed in a 14-pin package. The two timers (A and B) share the same power supply pins.

Low power versions of the 555 are made, such as the ICM7555, but these should only be used when specified (to increase battery life) because their maximum output current of about 20mA (with 9V supply) is too low for many standard 555 circuits. The ICM7555 has the same pin arrangement as a standard 555.



For further information please see the page on [555 and 556 timer circuits](#).

Logic ICs (chips)

Logic ICs process **digital signals** and there are many devices, including **logic gates**, flip-flops, shift registers, counters and display drivers. They can be split into two groups according to their pin arrangements: the **4000 series** and the **74 series** which consists of various families such as the 74HC, 74HCT and 74LS.

For most new projects the 74HC family is the best choice. The older 4000 series is the only family which works with a supply voltage of more than 6V. The 74LS and 74HCT families require a 5V supply so they are not convenient for battery operation.

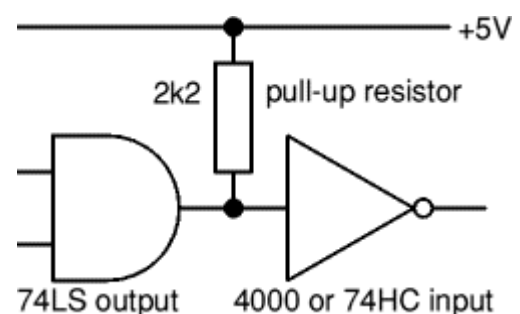
The table below summarises the important properties of the most popular logic families:

Property	4000 Series	74 Series 74HC	74 Series 74HCT	74 Series 74LS
Technology	CMOS	High-speed CMOS	High-speed CMOS TTL compatible	TTL Low-power Schottky
Power Supply	3 to 15V	2 to 6V	5V $\pm$ 0.5V	5V $\pm$ 0.25V
Inputs	Very high impedance. Unused inputs must be connected to +Vs or 0V. Inputs cannot be reliably driven by 74LS outputs unless a 'pull-up' resistor is used (see below).		Very high impedance. Unused inputs must be connected to +Vs or 0V. Compatible with 74LS (TTL) outputs.	'Float' high to logic 1 if unconnected. 1mA must be drawn out to hold them at logic 0.
Outputs	Can sink and source about 5mA (10mA with 9V supply), enough to light an LED. To switch larger currents use a transistor .	Can sink and source about 20mA, enough to light an LED. To switch larger currents use a transistor .	Can sink and source about 20mA, enough to light an LED. To switch larger currents use a transistor .	Can sink up to 16mA (enough to light an LED), but source only about 2mA. To switch larger currents use a transistor .
Fan-out	One output can drive up to 50 CMOS, 74HC or 74HCT inputs, but only one 74LS input.	One output can drive up to 50 CMOS, 74HC or 74HCT inputs, but only 10 74LS inputs.		One output can drive up to 10 74LS inputs or 50 74HCT inputs.
Maximum Frequency	about 1MHz	about 25MHz	about 25MHz	about 35MHz
Power consumption of the IC itself	A few μ W.	A few μ W.	A few μ W.	A few mW.

Mixing Logic Families

It is best to build a circuit using just one logic family, but if necessary the different families may be mixed providing the power supply is suitable for all of them. For example mixing 4000 and 74HC requires the power supply to be in the range 3 to 6V. A circuit which includes 74LS or 74HCT ICs must have a 5V supply.

A 74LS output cannot reliably drive a 4000 or 74HC input unless a 'pull-up' resistor of 2.2k Ω is



Driving 4000 or 74HC inputs from a 74LS output using a pull-up resistor.

connected between the +5V supply and the input to correct the slightly different logic voltage ranges used.

Note that a 4000 series output can drive only one 74LS input.

4000 Series CMOS

Quick links to individual ICs

This family of logic ICs is numbered from 4000 onwards, and from 4500 onwards. They have a B at the end of the number (e.g. 4001B) which refers to an improved design introduced some years ago. Most of them are in 14-pin or 16-pin packages. They use **CMOS circuitry** which means they use very little power and can tolerate a wide range of power supply voltages (3 to 15V) making them **ideal for battery powered projects**. CMOS is pronounced 'see-moss' and stands for Complementary Metal Oxide Semiconductor.

However the CMOS circuitry also means that they are [static sensitive](#). Touching a pin while charged with static electricity (from your clothes for example) may damage the IC. In fact most ICs in regular use are quite tolerant and earthing your hands by touching a metal water pipe or window frame before handling them will be adequate. ICs should be left in their protective packaging until you are ready to use them. For the more sensitive (and expensive!) ICs special equipment is available, including earthed wrist straps and earthed work surfaces.

[4000](#) [4060](#)
[4001](#) [4068](#)
[4002](#) [4069](#)
[4011](#) [4070](#)
[4012](#) [4071](#)
[4017](#) [4072](#)
[4020](#) [4073](#)
[4023](#) [4075](#)
[4024](#) [4077](#)
[4025](#) [4081](#)
[4026](#) [4082](#)
[4028](#) [4093](#)
[4029](#) [4510](#)
[4030](#) [4511](#)
[4040](#) [4516](#)
[4049](#) [4518](#)
[4050](#) [4520](#)

For further information, including pin connections, please use the quick links on the right or go to [4000 Series ICs](#).

74 Series: 74LS, 74HC and 74HCT

Quick links to individual ICs

There are several families of logic ICs numbered from 74xx00 onwards with letters (xx) in the middle of the number to indicate the type of circuitry, eg 74LS00 and 74HC00. The original family (now obsolete) had no letters, eg 7400.

The **74LS** (Low-power Schottky) family (like the original) uses TTL (Transistor-Transistor Logic) circuitry which is fast but requires more power than later families.

The **74HC** family has High-speed CMOS circuitry, combining the speed of TTL with the very low power consumption of the 4000 series. They are CMOS ICs with the same pin arrangements as the older 74LS family. Note that 74HC inputs cannot be reliably driven by 74LS outputs because the voltage ranges used for logic 0 are not quite compatible, use 74HCT instead.

The **74HCT** family is a special version of 74HC with 74LS TTL-compatible inputs so 74HCT can be safely mixed with 74LS in the same system. In fact 74HCT can be used as low-power direct replacements for the older 74LS ICs in most circuits. The minor disadvantage of 74HCT is a lower immunity to noise, but this is unlikely to be a problem in most situations.

[7400](#) [7432](#)
[7402](#) [7442](#)
[7403](#) [7447](#)
[7404](#) [7486](#)
[7405](#) [7490](#)
[7408](#) [7493](#)
[7409](#) [74132](#)
[7410](#) [74160](#)
[7411](#) [74161](#)
[7412](#) [74162](#)
[7414](#) [74163](#)
[7420](#) [74192](#)
[7421](#) [74193](#)
[7427](#) [74390](#)
[7430](#) [74393](#)

[74HC4017](#)
[74HC4020](#)
[74HC4040](#)
[74HC4060](#)
[74HC4511](#)

Beware that the 74 series is often still called the 'TTL series' even though the latest ICs do not use TTL!

For further information, including pin connections, please use the quick links on the right or go to [74 series ICs](#).

The CMOS circuitry used in the **74HC** and **74HCT** series ICs means that they are [static sensitive](#). Touching a pin while charged with static electricity (from your clothes for example) may damage the IC. In fact most ICs in regular use are quite tolerant and earthing your hands by touching a metal water pipe or window frame before handling them will be adequate. ICs should be left in their protective packaging until you are ready to use them.

PIC microcontrollers

A PIC is a **P**rogrammable **I**ntegrated **C**ircuit microcontroller, a 'computer-on-a-chip'. They have a processor and memory to run a program responding to inputs and controlling outputs, so they can easily achieve complex functions which would require several conventional ICs.

Programming a PIC microcontroller may seem daunting to a beginner but there are a number of systems designed to make this easy. The PICAXE system is an excellent example because it uses a standard computer to program (and re-program) the PICs; no specialist equipment is required other than a low-cost download lead. Programs can be written in a simple version of BASIC or using a flowchart. The PICAXE programming software and extensive documentation is available to download free of charge, making the system ideal for education and users at home. For further information (including downloads) please see www.picaxe.co.uk



www.picaxe.co.uk

If you think PICs are not for you because you have never written a computer program, please look at the PICAXE system! It is very easy to get started using a few simple BASIC commands and there are a number of projects available as kits which are ideal for beginners. The system is stocked by [Rapid Electronics](#).

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